Supplementary Material

From Aesthetics to Learning: A Bibliometric and Conceptual Review of User Experience Design in Digital Learning Platforms (2015-2025)

1. Access to the Supporting Data

Supplementary workbook (Google Sheets, view-only): availability

The link to the workbook is not included in this document. It will be provided, if needed, once the review process has been completed.

This document is self-contained: the coding rules, the corpus statistics, the evidence funnel, the sensitivity analysis, and the validation results are reported in Sections 2 to 6. Once released, the workbook will contain the derived, record-level coding output and the summary sheets behind every figure in the manuscript. To respect the terms of the source database (Scopus) and to keep the material anonymous, it will not reproduce titles, abstracts, or keywords of the 6,404 analysed records. The rule set that generated each code is summarised in Section 3 and will be given in full in the sheet Coding_Dictionary.

Table 1. Planned contents of the supplementary workbook

Sheet	Content	Supports (manuscript)
Records_Flags	One row per record: ID, year, citation count, primary domain, and 0/1 flags for every coded construct (technology, pedagogy, 22 UX attributes, five evidence stages, strict/broad variants, false-positive flag).	All results
Coding_Dictionary	Complete regular-expression patterns for every rule, exactly as applied.	Section III.B-C
Summary_RQ1	Domain distribution of the analytic corpus (N = 6,404).	IV.A
RQ1_TechPedagogy_Index	Technology- vs pedagogy-oriented share per year.	IV.C, Fig. 3
RQ2a_22Attr_Heatmap	Counts and shares of 22 UX attributes, app vs user aspect.	IV.D, Fig. 4
RQ2b_Heatmap_Top10AttrXDomain	Within-domain prevalence of the top-10 UX attributes.	IV.E, Fig. 5
RQ3_Evidence_Architecture	Five-stage cumulative evidence funnel.	IV.F, Fig. 6
RQ3_Sensitivity_Analysis	Strict/original/broad keyword definitions × full corpus, top 10%, top 5% most-cited.	IV.F
RQ3_Validation_Summary, RQ3_TopCited_SpotCheck, RQ3_StratValidation_DomYear	Automated vs manual coding for 212 records (title and coding columns only), with per-stage TP/FP/FN/TN, precision, recall, F1.	III.C, IV.F

Sheet names use the analysis workbook's internal numbering (RQ1-RQ3), which maps to the manuscript's three research questions. Figs. 1-2 (keyword co-occurrence network, topic trends) were produced with Biblioshiny/bibliometrix from the same export (top 50 keywords, minimum 20 occurrences, Louvain clustering, PageRank node size).

2. Data Provenance and Corpus Cleaning

- **Source:** Scopus, data collection closed on 31 December 2025; English-language journal articles and conference papers published 2015-2025.
- **Search string:** ("user experience" OR "user interface" OR "usability") AND ("e-learning" OR "online learning" OR "digital learning" OR "learning platform" OR "learning management system").
- **Cleaning:** 6,697 processed records; 293 (4.38%) were flagged by a rule-based off-topic filter (e.g., federated learning, learning-to-rank, network anomaly detection, histopathology) unless learner-context terms were present, leaving the analytic corpus of N = 6,404. Flagged records will be retained in the workbook with Likely_FalsePositive = 1 so the exclusion can be audited.
- **Robustness to cleaning:** the technology-oriented share is 91.06% (full) vs 90.99% (clean) and the pedagogy-oriented share 27.36% vs 27.56%, differences below 0.3 percentage points.

- **Growth:** the 11.87% average annual growth reported in the manuscript was computed by Biblioshiny on the processed export (335 records in 2015 to 1,028 in 2025); on the analytic corpus (328 to 961) it is 11.35%.

3. Rule-Based Coding Scheme

Every classification is a case-insensitive pattern match on one field, SearchText = lower-case (Title + Abstract + Author Keywords + Index Keywords). The same rules are applied uniformly to all records; no per-record manual judgement enters the counts. Manual coding was used only to validate the automated rules (Section 5.3).

Table 2. Primary-domain rules (single label; first matching rule wins) and resulting distribution

Priority	Domain	Trigger terms (abridged)	n	%
1	Corporate training	corporate / workplace / employee / professional / staff training	46	0.72
2	Blended learning	blended, hybrid learning, flipped classroom	170	2.65
3	MOOC	MOOC, massive open online course	174	2.72
4	Mobile learning	mobile learning, m-learning, mobile app, mobile device	726	11.34
5	LMS / VLE	learning management system, LMS, Moodle, Canvas, Blackboard, VLE	707	11.04
6	VR/AR & gamification	virtual / augmented reality, immersive, gamification, game-based learning, serious game	1,592	24.86
7	AI & adaptive	artificial intelligence, machine / deep learning, adaptive learning, chatbot, generative AI, intelligent tutoring	667	10.42
8	Medical / health education	medical, nursing, clinical, health care education; medical student	106	1.66
9	Language learning	language learning, English learning, second language, EFL, ESL	51	0.80
10	Formal education	K-12, elementary / primary / secondary school, higher education, university / college student, undergraduate	426	6.65
11	General e-learning	e-learning, online learning, online course / platform, distance learning / education	1,439	22.47
12	Other / niche	no rule above matched	300	4.68
	Total		**6,404**	**100.00**

Source sheet: Summary_RQ1. Percentages are shares of the analytic corpus (N = 6,404).

Table 3. Evidence-chain rules

Stage	Operational rule	Workbook column
1. Design Feature	At least one of four interface-design constructs: visual design, interface design, content organisation, functionality.	DesignFeature_V8
2. UX Response	Pragmatic (usability, usability testing, heuristic evaluation, system usability) OR experiential (UX-evaluation phrasing, satisfaction, engagement, immersion, emotional, hedonic terms).	PragmaticUX_Signal, ExperientialUX_Signal
3. Behavioral Interaction	usability test, eye-tracking, A/B testing, learning analytics, task analysis, log data, clickstream.	Behavioral_EvidenceSignal
4. Learning Process	Explicit cognitive or self-regulatory construct (term list in Table 4).	LearningProcess_Signal
5. Learning Outcome	Measured learning result (term list in Table 4).	LearningOutcome_Signal

Logic is cumulative: a record counts at stage k only if it also satisfies stages 1 to k−1.

Table 4. Keyword variants used in the sensitivity analysis

List	Terms (each list extends the previous one)
Learning Process: Strict (7)	cognitive load; scaffold*; metacognit*; self-regulat*; knowledge construction; formative feedback; problem-solving process
Learning Process: Original (9)	+ learning process; information processing
Learning Process: Broad (15)	+ cognitive process; zone of proximal development; self-regulated learning; reflective practice; active learning process; meta-cognit*

List	Terms (each list extends the previous one)
Learning Outcome: Strict (15)	learning outcome; academic performance; academic achievement; test score; exam score; pre-test; post-test; pretest; posttest; knowledge gain; learning gain; GPA; skill acquisition; competency/competence achievement or gain; academic score
Learning Outcome: Original (19)	+ grade(s); learning effectiveness; learning performance; retention rate
Learning Outcome: Broad (28)	+ student performance; learner performance; academic result; learning effect; knowledge retention; exam result; quiz score; achievement score; learning improvement

4. Descriptive Results (RQ1 and RQ2)

Table 5. Technology- vs pedagogy-oriented publications per year

Year	Papers	Technology n	Technology %	Pedagogy n	Pedagogy %
2015	328	313	95.4	95	29.0
2016	287	253	88.2	82	28.6
2017	352	321	91.2	88	25.0
2018	395	364	92.2	103	26.1
2019	495	447	90.3	106	21.4
2020	577	498	86.3	147	25.5
2021	663	588	88.7	174	26.2
2022	727	663	91.2	173	23.8
2023	815	755	92.6	203	24.9
2024	804	736	91.5	218	27.1
2025*	961	889	92.5	376	39.1
Overall	**6,404**	**5,827**	**91.0**	**1,765**	**27.6**

*Source sheet: RQ1_TechPedagogy_Index. The two categories are not mutually exclusive (a record may contain both kinds of keywords). *2025 indexing was incomplete at data collection.*

Table 6. Ten most frequent UX attributes (multi-label; share of the 6,404 records)

Attribute	Orientation	Records	% of records	Rank
Usability	App (system) aspect	3,395	53.0	1
Effectiveness	App aspect	1,164	18.2	2
Engagement	User (learner) aspect	1,025	16.0	3
Efficiency	App aspect	976	15.2	4
Satisfaction	User aspect	843	13.2	5
Ease of use	App aspect	588	9.2	6
Emotional	User aspect	511	8.0	7
Novelty	App aspect	475	7.4	8
Usefulness	App aspect	352	5.5	9
Functionality	App aspect	294	4.6	10

Source sheet: RQ2a_22Attr_Heatmap (22 attributes, grouped into app vs user aspect following Alshammare et al., 2025). Across all 22 attributes there are 10,560 attribute mentions: 7,747 app-aspect (73.36%) and 2,813 user-aspect (26.64%). Within-domain prevalence of the top-10 attributes (e.g., usability from 39.3% in AI & adaptive to 77.4% in medical education; emotional UX at 28.1% in LMS/VLE) is in RQ2b_Heatmap_Top10AttrXDomain.

5. Evidence Chain, Sensitivity, and Validation (RQ3)

5.1 Cumulative evidence funnel

Table 7. UX-to-learning evidence funnel (N = 6,404)

Stage	Individual n	Individual %	Cumulative n	Cumulative %
1. Design Feature	3,387	52.89	3,387	52.89
2. UX Response	4,985	77.84	2,819	44.02
3. Behavioral Interaction	660	10.31	329	5.14
4. Learning Process	788	12.30	57	0.89
5. Learning Outcome	**901**	**14.07**	**14**	**0.22**

Source sheet: RQ3_Evidence_Architecture. Individual = records meeting the stage alone; cumulative = records meeting all stages up to and including it.

5.2 Sensitivity to keyword definitions and citation level

Table 8. Full-chain integration under alternative definitions

Subset	Keyword list	n	Behavioral (cum.)	Learning Process (cum.)	Full chain (cum.)
All records (clean)	Strict	6,404	329	23 (0.36%)	4 (0.06%)
All records (clean)	**Original**	**6,404**	**329**	**57 (0.89%)**	**14 (0.22%)**
All records (clean)	Broad	6,404	329	60 (0.94%)	17 (0.27%)
Top 10% most-cited	Strict	640	26	3 (0.47%)	0 (0.00%)
Top 10% most-cited	Original	640	26	5 (0.78%)	1 (0.16%)
Top 10% most-cited	Broad	640	26	5 (0.78%)	1 (0.16%)
Top 5% most-cited	Strict	320	16	3 (0.94%)	0 (0.00%)
Top 5% most-cited	Original	320	16	5 (1.56%)	1 (0.31%)
Top 5% most-cited	Broad	320	16	5 (1.56%)	1 (0.31%)

Subsets are the 640 and 320 records of the analytic corpus with the highest citation counts. Percentages are shares of the subset. Across all nine conditions the full-chain share stays between 0% and 0.31%.

5.3 Validation of automated coding against manual coding

Automated codes were compared with manual coding of 212 records by one human coder: Set A (n = 112) comprises the 100 most-cited records plus 12 records with a full-chain automated signal; Set B (n = 100) is a sample stratified by primary domain and year period (2015-2018, 2019-2021, 2022-2023, 2024-2025). Manual notes in the validation sheets record corrections made after re-reading the full text in ambiguous cases. The UX Response stage is validated through its two components, pragmatic and experiential.

Table 9. Agreement between automated and manual coding

Stage	Agr. A	F1 A	Agr. B	F1 B	Prec. A+B	Rec. A+B	F1 A+B	Agr. A+B
Design Feature	0.85	0.90	0.74	0.83	0.89	0.85	0.87	0.80
Pragmatic UX	0.95	0.95	0.98	0.98	1.00	0.93	0.97	0.96
Experiential UX	0.81	0.82	0.79	0.78	0.76	0.85	0.80	0.80
Behavioral Data	0.94	0.84	0.97	0.88	0.94	0.79	0.86	0.95
Learning Process	0.96	0.93	0.99	0.95	0.95	0.92	0.94	0.98
Learning Outcome	0.92	0.86	0.96	0.78	0.85	0.83	0.84	0.94
Mean over stages							**0.88**	**0.90**

Source sheets: RQ3_Validation_Summary (sections 1-3), RQ3_TopCited_SpotCheck, RQ3_StratValidation_DomYear. Agr. = agreement (accuracy) between automated and manual codes; A = Set A (n = 112), B = Set B (n = 100), A+B = pooled 212 records. The bottom row averages the six coded categories of the pooled set.

6. Where to Verify Each Headline Figure After Release

Table 10. Headline figures and their location in the workbook

Figure reported in the manuscript	Value	Workbook location
Analytic corpus size	6,404	Summary_RQ1, cell F3
Technology-oriented publications (mean)	91.0%	RQ1_TechPedagogy_Index, cell E16
Pedagogy-oriented publications (mean)	27.6%	RQ1_TechPedagogy_Index, cell F16
Usability as UX dimension	3,395 (53.0%)	RQ2a_22Attr_Heatmap, row Usability
App-aspect vs user-aspect share	73.36% / 26.64%	RQ2a_22Attr_Heatmap, rows “App aspect total” and “User aspect total” (share = row / sum of both)
Any behavioral method (independent)	660 (10.3%)	RQ3_Evidence_Architecture, row Behavioral Interaction, columns C and E
UX Response → Behavioral (cumulative)	44.02% → 5.14%	RQ3_Evidence_Architecture, columns D and F
Design feature → learning outcome	14 records (0.22%)	RQ3_Evidence_Architecture, row Learning Outcome, columns D and F
Sensitivity range of full-chain share	0.06%-0.27% (all); 0%-0.16% (top 10%); 0%-0.31% (top 5%)	RQ3_Sensitivity_Analysis
Validation accuracy / F1	0.90 / 0.88	RQ3_Validation_Summary, section 3

7. Notes on Objectivity and Limitations

- The counts are produced by transparent, fixed rules applied identically to every record, so they can be reproduced from the rule set (Section 3) and, once released, from the record-level flags, without relying on the authors’ judgement at analysis time.
- The choice of keywords is a researcher decision. It is therefore examined through strict, original, and broad definitions and through citation-based subsets (Table 8), and through comparison with manual coding (Table 9).
- Coding operates on titles, abstracts, and keywords rather than full texts, and validation relied on a single human coder; larger double-coded samples would strengthen it. A study may report a learning outcome in its full text without mentioning it in the abstract, so the evidence-chain shares should be read as visibility in the indexed record.